Maquette Logging Analysis

All the request send to FraudCheck API were logged onto maquette.fraudchek.audit* kibana dashboard. Logs contain a JSON object with all the information such as rules passes and their result.

Two types of logs are generated by system.

Audit logs - contains information of all the rule execution as JSON

Application logs - contains application logs like info, error, warn.

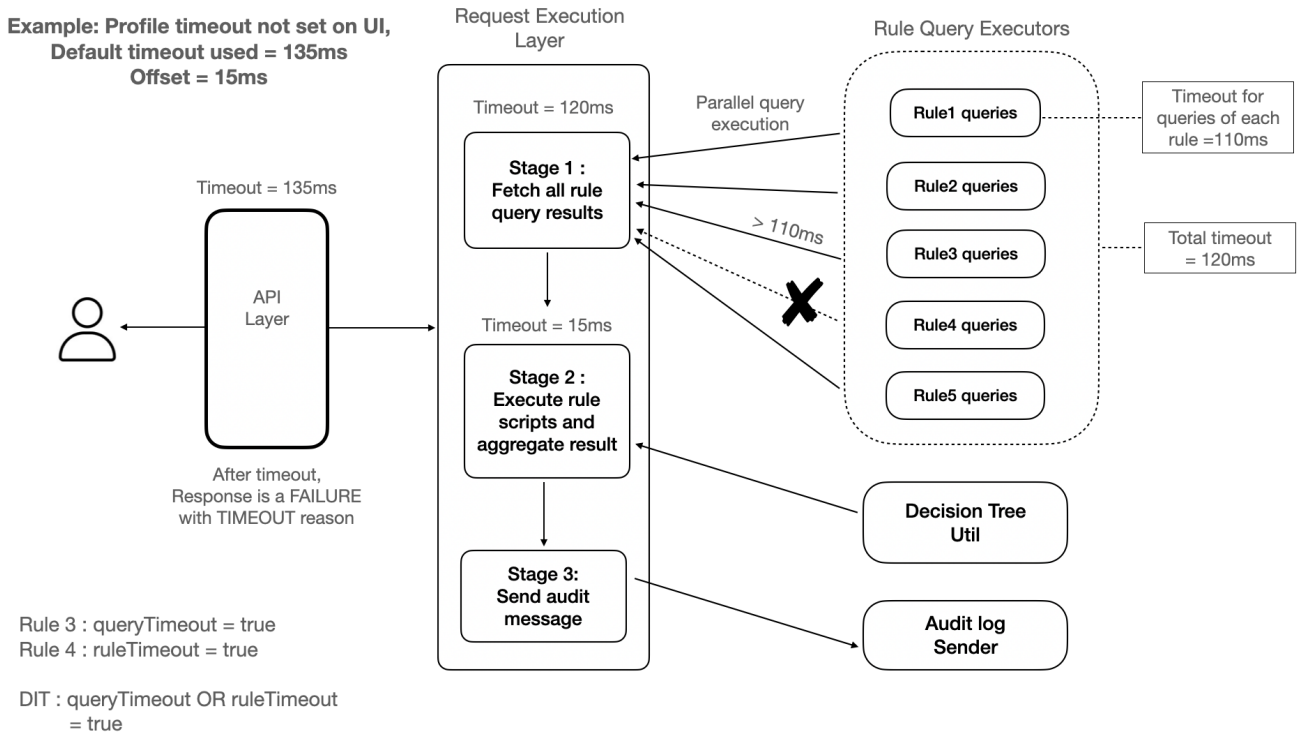
if audit log Entry not present in S3

- Kafka or consumer problem - kafka not available or log stash not table to keep up with produced rate of messages.
- For rule engine - Timeout happened before execution of ruleset - currently system produces msgs once rules start executing
- For rule admin, rule engine - System didn't receive request

if audit log Entry not present in ES

- Similar as above
- Schema has same key with conflicting format
  - merchantId as String for domain A vs Int for domain B - ES cannot infer the type in this case. These logs will be present on S3 but not on ES. Work around - Standardize the keys with client or use different keys **FRAUD-7584** - Authenticate to see issue details
- Number of indexed supported keys has been exceeded - happens when client passes a huge payload more than what is needed for rule execution. Reduce the number of keys in the request from client.
  - 1000 by default

Currently there is no record for the request which get timeout(before the execution of rule set. it might happen system is at overcapacity and request get timeout before executing any operation). We cannot see this in audit log but in the dashboard the timeout will increase. we can get the count of such request in promethues. https://docs.google.com/presentation/d/1tajkoWYkdYEZg5p78uETM7oKFEMDg4Ad5_FkbSvegKM/edit#slide=id.g245f1181fc5_0_22



If the API layer does not receive any response in 135ms then we get Failure in response and reason is timeout. It is this case which do not get into audit log.

RequestTimeout - It is set as true if the API deadline is overdue. This is set in Stage 3 of Request Execution layer.
Before sending the message to audit log sender, if the request has crossed the API timeout, requestTimeout is set to true.
It has no impact on DecisionImpactingTimeout. In the diagram, the timeout value is 135ms.

DecisionImpactingTimeout (DIT) - If any of the individual rule results contain DIT = true, the final DIT = true. This is irrespective of the rule status (LIVE /MONITOR). The individual DIT is calculated as described below.

The summary for each rule in stored in the actions array.

- For each action (rule summary), there are the following timeouts:

  - **QueryTimeout** - Each rule gets a fixed time to execute all its queries. If it crosses that time, it results in queryTimeout = true.
    In the above diagram, the each query executor timeout is 110ms. Rule3 will have *queryTimeout = true*.

  - **RuleTimeout** - Queries for all rules are run in parallel. If the query results of a rule were not returned to request executor before Stage 1 timeout, ruleTimeout is set to true.
    This could happen because of thread starvation, or delays in context switching, etc.
    In the above diagram, request executor did not receive Rule4 query results, within the 120ms timeout. Rule4 here will have *ruleTimeout = true*.

  - **DecisionImpactingTimeout** - If either QueryTimeout is true or ruleTimeout is true.

  - **ScriptTimeout** - The intention is to catch rule scripts that take over a fixed set of time and set their scriptTimeout to true This could happen if there are unintended infinite loops, etc.
    This field can be ignored for now. The business logic to set scriptTimeout needs improvement as of now.

  - **Timeout** - ruleTimeout = true OR queryTimeout = true OR scriptTimeout = true

Audit Logs: https://maquette-kibana.internal.ap-south-1.staging.osmose.risk.pai.mypaytm.com

- maquette.admin.* - logs, error
- maquette.feature.audit* - no logs
- **maquette.fraudcheck.audit* - logs**
- maquette.fraudcheck.audit.-critical-test-2023.05.18 - no logs
- maquette.fraudcheck.audit.-critical-test-2023.05.18 - no logs
- maquette.fraudcheck.audit.-noncritical-test* - no logs
- maquette.fraudcheck.audit.-noncritical-test* - no logs
- maquette.metrics.aggregate.rule* - no logs
- maquette.rule.actions* - logs

Application Logs: https://kibana.internal.ap-south-1.staging.osmose.risk.pai.mypaytm.com

- **pai-risk-osmose-staging-* - logs**